FACTORIAL STAIRCASES: A PATTERN IN FACTORIALS, EXPLAINED

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Abstract. This expository note starts from a pattern found by experiment. The expression $(n-m)\big[(n-1)!+(n-2)!+\cdots+(n-m-1)!\big]$ equals $n!$ when $m=1$ but falls short of $n!$ when $m\geq 2$, and the shortfall is always $(m-1)(n-1)!+(m-2)(n-2)!+\cdots+1\cdot(n-m+1)!$. We prove the pattern and explain it with the factorial number system. The shortfall is the number of permutations of $1,\ldots,n$ that come before, in dictionary order, the permutation that reverses the first m entries. Changing the multiplier leads to Kurepa's left factorial $!n=0!+1!+\cdots+(n-1)!$. Modulo a prime p, the bracket (with an obvious factorial removed) depends only on n modulo p. Its value is an alternating sum of falling factorials. When $m-n$ is congruent modulo p to some $j\leq m$ (for example when $n=p$, the case relevant to Kurepa's conjecture), this sum equals $(-1)^j d_j$, where d_j is a derangement number; otherwise it is a truncation of that sum. This is a direct extension of a lemma of Kellner. Nothing here needs more than telescoping sums, Lehmer codes and Wilson's theorem. The aim is to show how a simple observation leads to these classical ideas, not to present new theorems.

1. Introduction

Since $n!=n\cdot(n-1)!$ and $n=(n-1)+1$, we can write

$$n!=(n-1)\big[(n-1)!+(n-2)!\big]. \tag{1}$$

For $n=7$ this reads $5040=6\cdot(720+120)$. A natural guess for the next formula of this kind is to lower the multiplier by one and add one more factorial to the bracket:

$$(n-2)\big[(n-1)!+(n-2)!+(n-3)!\big].$$

This is not equal to $n!$. For $n=7$ it equals $5\cdot(720+120+24)=4320$, which is 720 less than $7!$. The missing amount, 720, is exactly $6!=(n-1)!$, and the same happens for every $n\geq 3$. Continuing in the same way gives Table 1. The shortfall is always a "staircase": a sum of factorials whose coefficients go down by one.

m	multiplier	shortfall $n!-(\text{multiplier})\cdot\big[(n-1)!+\cdots+(n-m-1)!\big]$
1	$n-1$	0
2	$n-2$	$(n-1)!$
3	$n-3$	$2(n-1)!+(n-2)!$
4	$n-4$	$3(n-1)!+2(n-2)!+(n-3)!$
5	$n-5$	$4(n-1)!+3(n-2)!+2(n-3)!+(n-4)!$

Table 1. The shortfall for the first five values of m. The bracket contains $m+1$ factorials.

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Notation. For integers $0 \le m \le n-1$ we write

$$L_m(n) = \sum_{i=1}^{m+1} (n-i)! = (n-1)! + (n-2)! + \cdots + (n-m-1)!$$

for the bracket, and

$$R_m(n) = n! - (n-m)\, L_m(n)$$

for the shortfall. The table suggests $R_m(n) = \sum_{i=1}^{m-1} (m-i)(n-i)!$.

Outline. Section 3 proves this formula for all m and n (Proposition 3.1). The proof is short, but it does not explain why the multiplier must be $n-m$, so the same section gives a second proof that does (Proposition 3.3). List all permutations of $1, 2, \ldots, n$ in dictionary order. Then $R_m(n)$ is the number of permutations that come before the permutation

$$w_m = m\,(m-1)\cdots 2\,1\,(m+1)\,(m+2)\cdots n,$$

which reverses the first m entries. The multiplier $n-m$ is the number of ways to move past w_m at each of its first m positions.

The rest of the note follows one thread, from this identity to a classical open problem. Each result is a short consequence of Lemma 2.1, of the telescoping argument in Proposition 3.1, or of Kellner's lemma; we include them as illustrations of these tools, not as new theorems.

- Section 4: any multiplier c with $1 \le c \le n-m$ gives a similar identity, attached to a different permutation. Taking $m = n-1$ brings in Kurepa's left factorial $!n = 0! + 1! + \cdots + (n-1)!$.
- Section 5: modulo a prime p, the bracket $L_m(n)$, after dividing out $(n-m-1)!$, depends only on n modulo p. Its value is a truncated alternating sum of falling factorials (Proposition 5.2). Let j be the residue of $m-n$ modulo p. If $j \le m$, which always holds when $n = p$, this sum is $(-1)^j d_j$, where d_j is a derangement number. If $j > m$ it is a truncation of that sum. The case $n = p$ is a lemma of Kellner, which restates Kurepa's conjecture.

Section 6 lists questions for further study. Appendix A collects side variations that use the same tools (other products such as double factorials, and a q-analogue). Appendix B describes the computer checks and the formal proofs.

What is known. All results in this note are elementary, and most are standard or close to standard.

- Proposition 3.1 is a rearrangement of the classical telescoping identity $\sum_{j=0}^{n-1} j \cdot j! = n! - 1$ [4].
- Lemma 2.1 is the factorial number system of Laisant [10] and Lehmer [11]; see also [8, §7.2.1.2]. Propositions 3.3 and 4.1 are direct applications of it, and Proposition A.1 is the same argument for a mixed-radix number system [7, 3].
- The numbers $R_m(n)$ already appear in the OEIS in two special cases. For $m = 3$, $R_3(n)$ is term $n-2$ of A052649 [15] (offset 0), where a 2004 comment by Eugene McDonnell notes the interpretation with three reversed entries. For $m = n-1$, $R_{n-1}(n) = n! - !n$ is term n of A065355 [16].
- Proposition A.3 is the usual refinement of a counting argument by inversions [20, Ch. 1].
- Proposition 5.2 is a short extension of [6, Lemma 2.4], which is the case $n = p$. Kellner restates Kurepa's conjecture [9, 5] as $p \nmid d_{p-1}$ in [6, Lemma 2.1 and Corollary 2.2]; see also Meštrović [12] and, for recent equivalent forms of the conjecture, Petojević et al. [19].

- The primes in Table 2 can be read off from the residues of $!p$ modulo p, which are tabulated in OEIS A100612 [17]. Its row $k = 2$ is A258944 [18].

We have not seen the general statements written down in exactly this form, but specialists would regard them as routine. For this reason we call them propositions rather than theorems. Every result was checked by computer, and most were also proved formally in the proof assistant Lean 4 with the Mathlib library. Appendix B lists what was checked and how; the scripts and Lean files are embedded in this PDF as attachments.

2. Permutations in dictionary order

Let $\mathfrak{S}_n$ be the set of permutations of $\{1, \dots, n\}$, written as words $w = w_1 w_2 \cdots w_n$. We compare two permutations the way a dictionary compares words. Look at the first position where they differ; the one with the smaller entry there comes first. We write $u <_{\text{lex}} w$ when u comes before w. The *rank* of w is the number of permutations that come before it:

$$\operatorname{rank}(w) = \#\{u \in \mathfrak{S}_n : u <_{\text{lex}} w\}.$$

The first permutation $12 \cdots n$ has rank 0, and the last one, $n \cdots 21$, has rank $n! - 1$.

There is a simple formula for the rank. For each position i, let

$$\ell_i(w) = \#\{j > i : w_j < w_i\}, \qquad g_i(w) = \#\{j > i : w_j > w_i\}.$$

These count the entries to the right of position i that are smaller and larger than w_i. Since there are $n - i$ entries to the right of position i, we have $\ell_i(w) + g_i(w) = n - i$. The list $\ell_1(w), \dots, \ell_n(w)$ is called the *Lehmer code* of w [11, 10].

Lemma 2.1 (factorial number system [11, 8])**.** *For every $w \in \mathfrak{S}_n$,*

$$\operatorname{rank}(w) = \sum_{i=1}^{n} \ell_i(w)\,(n-i)! \qquad \text{and} \qquad \#\{u \in \mathfrak{S}_n : w \le_{\text{lex}} u\} = 1 + \sum_{i=1}^{n} g_i(w)\,(n-i)!.$$

Proof. Take $u \neq w$, and let i be the first position where they differ. The entries still unused at position i are exactly $w_i, w_{i+1}, \dots, w_n$. If u_i is one of the $\ell_i(w)$ entries smaller than w_i, then u comes before w. If it is one of the $g_i(w)$ entries larger than w_i, then u comes after w. In either case the remaining $n - i$ positions can be filled in $(n - i)!$ ways. Summing over i gives both formulas; the extra 1 in the second formula counts w itself. □

Example 2.2. For $w = 3214$: the entry 3 has two smaller entries to its right, so $\ell_1 = 2$. The entry 2 has one, so $\ell_2 = 1$. Hence $\operatorname{rank}(3214) = 2 \cdot 3! + 1 \cdot 2! = 14$. Indeed, 3214 is the fifteenth permutation of $\{1, 2, 3, 4\}$ in dictionary order.

3. The staircase identity and what it counts

Proposition 3.1 (staircase identity; a form of $\sum j \cdot j! = n! - 1$)**.** *For all integers $0 \le m \le n-1$,*

$$n! = (n - m) \sum_{i=1}^{m+1} (n-i)! \; + \; \sum_{i=1}^{m-1} (m-i)\,(n-i)!.$$

In other words, $R_m(n) = \sum_{i=1}^{m-1} (m-i)(n-i)!$.

Proof. The only fact we need is $j \cdot j! = (j+1)! - j!$. Write the multiplier as $n - m = (n-i) - (m-i)$, and treat the two parts separately. First,

$$\sum_{i=1}^{m+1} (n-i)\,(n-i)! = \sum_{i=1}^{m+1} \big[(n-i+1)! - (n-i)!\big] = n! - (n-m-1)!,$$

because the sum telescopes. Second, in $\sum_{i=1}^{m+1}(m-i)(n-i)!$ the term $i=m$ is zero and the term $i=m+1$ is $-(n-m-1)!$, so

$$\sum_{i=1}^{m+1}(m-i)\,(n-i)! = \sum_{i=1}^{m-1}(m-i)\,(n-i)! - (n-m-1)!.$$

Subtracting the second equation from the first gives $(n-m)L_m(n) = n! - \sum_{i=1}^{m-1}(m-i)(n-i)!$. □

Remark 3.2. The proof uses only $j\cdot j! = (j+1)! - j!$, so it also works for the Gamma function, which satisfies $\Gamma(z+1) = z\,\Gamma(z)$. For an integer $m \geq 0$ and a complex number x that is not one of $m, m-1, m-2, \ldots,$

$$\Gamma(x+1) = (x-m)\sum_{i=1}^{m+1}\Gamma(x-i+1) + \sum_{i=1}^{m-1}(m-i)\,\Gamma(x-i+1),$$

by the same telescoping with $(x-i)\Gamma(x-i+1) = \Gamma(x-i+2) - \Gamma(x-i+1)$.

The proof works, but it does not explain where the multiplier $n-m$ comes from. The next proposition does.

Proposition 3.3 (for $m=3$, see [15])**.** *Let* $0 \leq m \leq n-1$ *and* $w_m = m\,(m-1)\cdots 1\,(m+1)\cdots n$. *Then*

$$\operatorname{rank}(w_m) = \sum_{i=1}^{m-1}(m-i)(n-i)! = R_m(n) \quad \textit{and} \quad \#\{u \in \mathfrak{S}_n : w_m \leq_{\text{lex}} u\} = (n-m)\sum_{i=1}^{m+1}(n-i)!.$$

So Proposition 3.1 says that the permutations before w_m *and the permutations from* w_m *on together make up all* $n!$ *permutations.*

Proof. For $i \leq m$ the entry of w_m in position i is $m-i+1$. All the smaller values $1, \ldots, m-i$ lie to its right, and so do all the $n-m$ values $m+1, \ldots, n$, which are larger. After position m the entries increase. So

$$\ell_i(w_m) = m-i, \quad g_i(w_m) = n-m \qquad (i \leq m),$$
$$\ell_i(w_m) = 0, \quad g_i(w_m) = n-i \qquad (i > m).$$

By Lemma 2.1, $\operatorname{rank}(w_m) = \sum_{i=1}^{m}(m-i)(n-i)!$, and the term $i=m$ is zero. For the second count, Lemma 2.1 gives

$$(n-m)\sum_{i=1}^{m}(n-i)! + \Big(1 + \sum_{i=m+1}^{n}(n-i)(n-i)!\Big) = (n-m)\sum_{i=1}^{m}(n-i)! + (n-m)!,$$

and $(n-m)! = (n-m)\,(n-m-1)!$ supplies the last term of the bracket. □

The proof shows the meaning of each part of the identity. Build a permutation from left to right, comparing it with w_m. At each of the first m positions, the available values split into smaller ones, which put the permutation before w_m, and larger ones, which put it after. There are $m-i$ smaller values at position i, and their contributions form the staircase. There are always exactly $n-m$ larger values, namely $m+1, \ldots, n$, and this constant is the multiplier. Figures 1 and 2 show this.

Example 3.4. For $n=4$ and $m=3$ we have $w_3 = 3214$. Proposition 3.3 says that $R_3(4) = 2\cdot 3! + 1\cdot 2! = 14$ permutations come before 3214, and $(4-3)(3!+2!+1!+0!) = 10$ permutations come from 3214 on; $14+10 = 24 = 4!$.

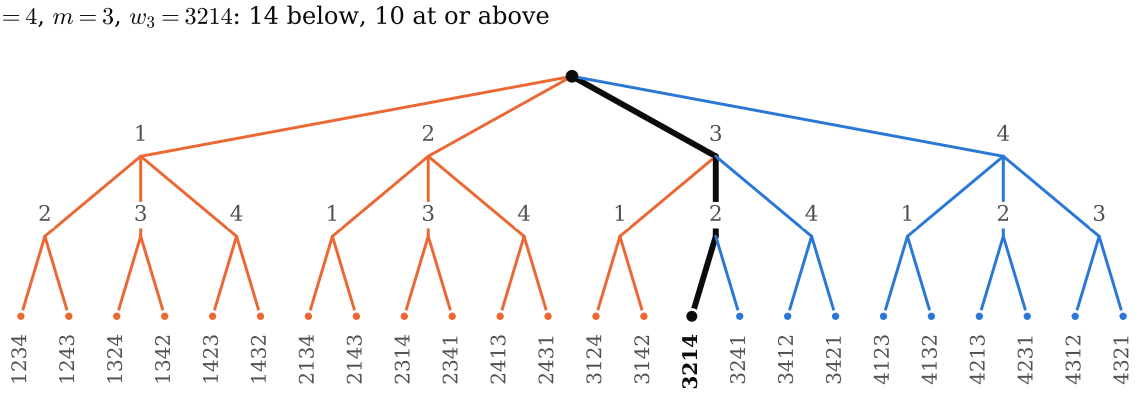


FIGURE 1. The 24 permutations of $\{1, 2, 3, 4\}$ as a tree, read from top to bottom; the leaves are in dictionary order. The black path leads to $w_3 = 3214$. Orange: the $R_3(4) = 14$ permutations before it. Blue: 3214 and the 9 permutations after it. At each of the first three levels, the black path has exactly $n - m = 1$ blue branch to its right.

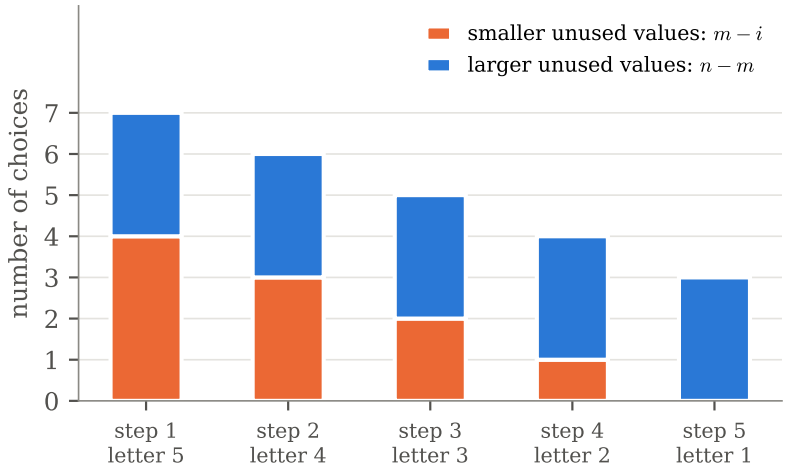


FIGURE 2. The choices at each of the first five positions of $w_5 = 54321678 \in \mathfrak{S}_8$. Orange: values smaller than the entry of w_5 $(4, 3, 2, 1, 0)$. Blue: larger values (always $n - m = 3$).

4. OTHER MULTIPLIERS

If we use a multiplier c smaller than $n - m$, we still get an identity, but it is attached to a different permutation.

Proposition 4.1. *Let $0 \le m \le n - 1$ and $1 \le c \le n - m$. Then*

$$n! = c \sum_{i=1}^{m+1} (n - i)! \; + \; \sum_{i=1}^{m} (n - i - c)(n - i)! \; + \; (n - m - c)(n - m - 1)!,$$

and the last two terms together are the rank of the permutation

$$w_{m,c} = (n - c)\,(n - c - 1) \cdots (n - c - m + 1) \;\; (n - c + 1) \;\; 1\,2 \cdots (n - c - m) \;\; (n - c + 2) \cdots n.$$

Proof. The word $w_{m,c}$ consists of four blocks: a decreasing block of m values, then the value $n-c+1$, then the values below the first block in increasing order, then the values above $n-c+1$ in increasing order. The conditions $1 \le c \le n-m$ make sure these are the values $1,\dots,n$, each used once.

Each of the first m entries has all smaller values to its right, since every entry to its left is larger. So $\ell_i = n-c-i$ for $i \le m$. The entry $n-c+1$ in position $m+1$ has $1,\dots,n-c-m$ to its right, so $\ell_{m+1} = n-m-c$. All later entries have $\ell_i = 0$. Lemma 2.1 gives the rank. The identity itself follows from

$$\sum_{i=1}^{m}(n-i)(n-i)! + (n-m)(n-m-1)! = \big(n! - (n-m)!\big) + (n-m)! = n!$$

by splitting $n-i = c + (n-i-c)$ and $n-m = c + (n-m-c)$. □

For $c = n-m$ the last term vanishes, $w_{m,c} = w_m$, and we recover Proposition 3.1. This is the only value of c for which the term with $(n-m-1)!$ disappears. That is why the pattern in Table 1 looks so simple.

Example 4.2. Take $n=5$, $m=2$, $c=1$. Then $w_{2,1} = 43\,5\,12$, and the proposition says $120 = 1\cdot(4!+3!+2!) + (3\cdot 4! + 2\cdot 3!) + 2\cdot 2! = 32 + 84 + 4$.

Corollary 4.3. *Let* $!n = 0! + 1! + \cdots + (n-1)!$ *be Kurepa's left factorial* [9, 14]. *Then*

$$n! = !n + \sum_{j=1}^{n-1}(j-1)\,j!,$$

and $!n$ *is the number of permutations from* $(n-1)(n-2)\cdots 1\,n$ *onwards.*

Proof. Take $m = n-1$. Then $1 \le c \le n-m$ forces $c=1$, and the bracket becomes $\sum_{i=1}^{n}(n-i)! = !n$. □

5. The bracket modulo a prime

At the end of the family is Kurepa's left factorial $!n = L_{n-1}(n)$. Kurepa conjectured that $!p$ is not divisible by p for any odd prime p [9] (see also [5, B44]). The conjecture is still open: computer searches have found no counterexample with $p < 2^{40}$ [1], and recent papers continue to study equivalent forms of it [19]. This suggests asking, for every m and n, what $L_m(n)$ is modulo a prime. This section answers that question for every prime.

5.1. **Removing the obvious factor.** Every term of $L_m(n) = (n-1)! + \cdots + (n-m-1)!$ is a multiple of the smallest one, $(n-m-1)!$. Dividing it out,

$$L_m(n) = (n-m-1)!\,P_m(n), \tag{2}$$

where, with $a = n-m$,

$$P_m(n) = 1 + a + a(a+1) + \cdots + a(a+1)\cdots(a+m-1).$$

For example, $P_2(n) = 1+(n-2)+(n-2)(n-1) = (n-1)^2$. We write $a^{\overline{k}} = a(a+1)\cdots(a+k-1)$ and $j^{\underline{k}} = j(j-1)\cdots(j-k+1)$, both equal to 1 for $k=0$, so that $P_m(n) = \sum_{k=0}^{m} a^{\overline{k}}$.

In the case that matters for Kurepa's conjecture, the answer involves the *derangement numbers* d_j [20, §2.2], [13]. These count the permutations of j objects in which no object stays in its place:

$$d_j = j!\sum_{i=0}^{j}\frac{(-1)^i}{i!}, \qquad (d_0, d_1, d_2, \dots, d_6) = (1, 0, 1, 2, 9, 44, 265).$$

5.2. **Periodicity modulo p.** Modulo p, a product of consecutive numbers starting at a looks like a product of consecutive numbers counting down from j, where j is the remainder of $-a$.

Lemma 5.1. *Let p be prime, and let $a \equiv -j \pmod p$ with $0 \le j \le p-1$. Then $a^{\overline{k}} \equiv (-1)^k j^{\underline{k}} \pmod p$ for all $k \ge 0$. In particular $a^{\overline{k}} \equiv 0$ when $k > j$.*

Proof. Each factor satisfies $a+t \equiv -(j-t) \pmod p$. If $k > j$, the factor with $t = j$ is divisible by p. □

Proposition 5.2 (extends [6])**.** *Let p be prime and $0 \le m \le n-1$. Let j be the number in $\{0, 1, \ldots, p-1\}$ with $j \equiv m - n \pmod p$. Then*

$$P_m(n) \equiv \sum_{k=0}^{\min(m,j)} (-1)^k \, j^{\underline{k}} \pmod p.$$

If $j \le m$, this is $(-1)^j \, d_j$. In particular, for fixed m the residue of $P_m(n)$ modulo p depends only on n modulo p.

Proof. Apply Lemma 5.1 to each term of $P_m(n) = \sum_{k=0}^{m} a^{\overline{k}}$ with $a = n - m \equiv -j$. The terms with $k > j$ vanish. If $j \le m$, the sum runs over $0 \le k \le j$, and substituting $i = j - k$ gives

$$\sum_{k=0}^{j} (-1)^k \frac{j!}{(j-k)!} = (-1)^j \, j! \sum_{i=0}^{j} \frac{(-1)^i}{i!} = (-1)^j d_j.$$ □

Example 5.3. Take $m = 2$, so $P_2(n) = (n-1)^2$ and $j \equiv 2 - n$. The proposition predicts $P_2(n) \equiv 1$ when $n \equiv 2$ ($j = 0$, $d_0 = 1$), $\equiv 0$ when $n \equiv 1$ ($j = 1$, $d_1 = 0$), and $\equiv 1$ when $n \equiv 0$ ($j = 2$, $d_2 = 1$). Indeed $(n-1)^2$ is 1, 0 and 1 modulo p in these three cases.

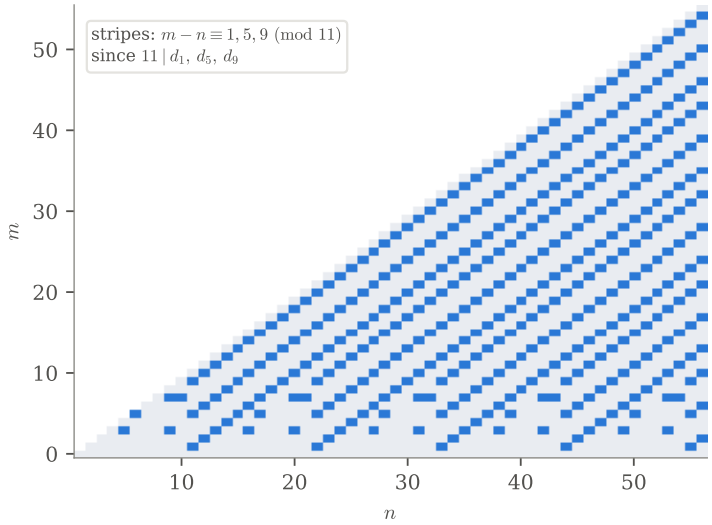


FIGURE 3. The pairs (n, m) for which 11 divides $P_m(n)$ (blue). By Proposition 5.2 they lie on the diagonals $m - n \equiv 1, 5, 9 \pmod{11}$, because 11 divides $d_1 = 0$, $d_5 = 44$ and $d_9 = 133496$. Near the bottom edge, where $j > m$, the shortened sum applies instead.

5.3. **The case $n = p$: Kellner's lemma and Kurepa's conjecture.**

Corollary 5.4 ([6, Lemma 2.4 and Lemma 2.1])**.** *Let p be prime and $0 \le m \le p-1$. Then*

$$(p-1)! + (p-2)! + \cdots + (p-m-1)! \;\equiv\; -\frac{d_m}{m!} \pmod p.$$

In particular $!p \equiv d_{p-1} \pmod p$, so Kurepa's conjecture is equivalent to the statement that p does not divide d_{p-1} for any odd prime p.

Proof. For $n = p$ we have $j = m$, so $P_m(p) \equiv (-1)^m d_m$ by Proposition 5.2. Wilson's theorem says $(p-1)! \equiv -1$. Since $(p-1)! = (p-m-1)!\,(p-m)\cdots(p-1) \equiv (p-m-1)!\,(-1)^m m!$, we get $(p-m-1)! \equiv (-1)^{m+1}/m!$, and multiplying gives the first claim. For $m = p-1$ the right side is $-d_{p-1}/(p-1)! \equiv d_{p-1}$. □

Example 5.5. For $p = 11$ and $m = 5$, $d_5 = 44$ is divisible by 11, so $10! + 9! + 8! + 7! + 6! + 5!$ is divisible by 11.

Kellner states the first claim as $K(p) - K(p-l) \equiv -S(l-1)/(l-1)!$ [6, Lemma 2.4], where K is the left factorial and S the derangement number; the second is his Lemma 2.1 for $n = p$. Proposition 5.2 shows that this congruence is one value of a periodic pattern that holds for all n. For example, taking $m = n-1 \ge p-1$ gives $j = p-1$, and hence $!n \equiv d_{p-1} \pmod p$ for every $n \ge p$. Kellner does not state this directly, but it follows from his Lemma 2.1 together with his Theorem 3.3, which says that $!n$ modulo p is constant for $n \ge p$.

5.4. **Common factors with $n!$.** Kurepa's conjecture is often stated as $\gcd(!n, n!) = 2$ for $n \ge 2$. The next corollary describes $\gcd\big(L_m(n), n!\big)$ for every m.

Corollary 5.6. *Let $0 \le m \le n-1$ and $G_m(n) = \gcd\big(L_m(n), n!\big)/(n-m-1)!$. Then*

$$G_m(n) = \gcd\big(P_m(n),\ (n-m)(n-m+1)\cdots n\big).$$

A prime p divides $G_m(n)$ if and only if both of the following hold:

1. *some multiple of p lies among $n-m, \dots, n$;*
2. *the sum in Proposition 5.2 is divisible by p (when $j \le m$, this means $p \mid d_j$).*

Moreover, $n-m+1$ always divides $G_m(n)$ when $m \ge 1$.

Proof. Both $L_m(n) = (n-m-1)!\,P_m(n)$ and $n! = (n-m-1)!\,(n-m)\cdots n$ contain the factor $(n-m-1)!$, which gives the formula for $G_m(n)$. A prime divides the product $(n-m)\cdots n$ exactly when (1) holds, and divides $P_m(n)$ exactly when (2) holds, by Proposition 5.2. For the last claim, the two smallest terms of $L_m(n)$ add up to $(n-m)! + (n-m-1)! = (n-m-1)!\,(n-m+1)$, and every other term is a multiple of $(n-m+1)!$. So $n-m+1$ divides $P_m(n)$, and it is one of the numbers $n-m, \dots, n$. □

For $m = n-1$ this says $G_{n-1}(n) = \gcd(!n, n!)$ is divisible by 2, and Kurepa's conjecture is that it equals 2. More generally, we may ask when $G_m(n)$ is as small as possible, namely equal to $n-m+1$. This happens for 862 of the 1710 pairs with $3 \le n < 60$ and $1 \le m \le n-1$; the corollary names the primes responsible in the other cases.

5.5. **Near-Kurepa primes.** Taking $m = p-1-k$ in Corollary 5.4 and simplifying with Wilson's theorem gives

$$!p - !k \;\equiv\; (-1)^k\, k!\, d_{p-1-k} \pmod p \qquad (0 \le k < p).$$

So a prime $p > k$ divides $!p - !k$ exactly when it divides d_{p-1-k}. The case $k = 0$ with p odd is Kurepa's question ($p = 2$ divides $!2 = 2$ trivially), for which no example exists with $p < 2^{40}$ [2, 1]. For $k \ge 1$, examples exist (Table 2).

k	$!k$	odd primes $p < 2 \cdot 10^6$ with $p > k$ and $p \mid !p - !k$
0	0	none
1	1	3, 11
2	2	31, 373
3	4	5, 41, 163
4	10	13, 151
5	34	7, 11
6	154	283, 1867, 4099
7	874	103, 2423, 254039, 304099
8	5914	42961, 1528781

TABLE 2. Odd primes dividing $!p - !k$, equivalently d_{p-1-k}. The prime $p = 2$ is left out; it divides $!2 - !0 = 2$ and no other entry. The row $k = 2$ is OEIS A258944 [18], whose next term 6855730873 was found by Andrejić and Tatarević [2].

The two primes in the row $k = 2$ multiply to $31 \cdot 373 = 11563$. Meštrović [12] showed that 11563 is a counterexample to his "strong Kurepa hypothesis" about a related determinant. His criterion for an odd composite n is $d_{n-1} \equiv 2 \pmod{n}$. For each prime p dividing n he shows $d_{n-1} \equiv d_{p-1} \pmod{p}$, so for $n = 31 \cdot 373$ the criterion becomes $p \mid d_{p-1} - 2$ for $p = 31, 373$; by the formula above this is $p \mid !p - !2$. We checked directly that $d_{11562} \equiv 2 \pmod{11563}$.

6. Questions for further study

Question 6.1. Are there similar identities for other orders on $\mathfrak{S}_n$, for example colexicographic order, or for reversing the *last* m entries?

Question 6.2. Corollary 5.6 says which primes divide $G_m(n)$, but not to which power. What is the exact power of p dividing $P_m(n)$? What proportion of pairs (m, n) have $G_m(n) = n - m + 1$ as n grows?

Question 6.3. For each fixed $k \geq 1$, are there infinitely many primes p with $p \mid !p - !k$? If the residues $d_{p-1-k} \bmod p$ behaved like random numbers, the number of such primes up to x would grow like $\log \log x$. That is slowly, but without bound, consistent with Table 2.

Question 6.4. Is there a version of Proposition A.3 for the major index instead of inversions? A computation shows that the two statistics are *not* equally distributed on $\{u : w_m \leq_{\text{lex}} u\}$ when $2 \leq m \leq n - 1$ and $n \leq 7$, so such a version would have a different form.

Question 6.5. Does Proposition 4.1 have a mixed-radix version, and for which multipliers is the remainder again an equally spaced staircase?

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Appendix A. Variations on the identity

This appendix collects two variations that use the same two tools, the telescoping sum and the ranking lemma. None of them is needed for the main text.

A.1. **Other products.** The proof of Proposition 3.1 used only one property of $n!$: it is obtained from $(n-1)!$ by multiplying by a number. Let $r_1, r_2, \dots$ be positive integers (the *growth factors*), and put $a_0 = 1$ and $a_k = r_1 r_2 \cdots r_k$. Factorials are the case $r_k = k$, double factorials $2 \cdot 4 \cdots (2k)$ the case $r_k = 2k$, and powers b^k the case $r_k = b$.

Permutations are replaced by digit strings. Let W_n be the set of strings $d_1 d_2 \cdots d_n$ with $0 \le d_i \le r_{n-i+1} - 1$, in dictionary order. These are the numerals of a mixed-radix number system, where each digit has its own base [7, §4.1], [3]; for $r_k = k$ the Lehmer code turns permutations into exactly these strings, in the same order.

Proposition A.1. *For $0 \le m \le n-1$,*

$$a_n = r_{n-m} \sum_{i=1}^{m+1} a_{n-i} \;+\; \sum_{i=1}^{m} \bigl(r_{n-i+1} - r_{n-m} - 1\bigr)\, a_{n-i}. \tag{3}$$

If $r_k > r_{n-m}$ for $n-m < k \le n$, all coefficients are nonnegative. Then the second sum is the number of strings before $t = t_1 \cdots t_m 0 \cdots 0$, where $t_i = r_{n-i+1} - r_{n-m} - 1$, and the first term is the number of strings from t on.

Proof. Since $a_k - a_{k-1} = (r_k - 1)a_{k-1}$, the sum $\sum_{i=1}^{m} (r_{n-i+1} - 1) a_{n-i}$ telescopes to $a_n - a_{n-m}$, and $a_{n-m} = r_{n-m} a_{n-m-1}$. Splitting $r_{n-i+1} - 1 = r_{n-m} + (r_{n-i+1} - r_{n-m} - 1)$ gives (3). For the count, argue as in Lemma 2.1: a string that first differs from t at position $i \le m$ has t_i smaller and r_{n-m} larger choices there, and a_{n-i} ways to fill the rest. □

Remark A.2. The staircase coefficient of a_{n-i} is $r_{n-i+1} - r_{n-m} - 1$, so the staircase goes down in equal steps exactly when the growth factors go up in equal steps. For $r_k = \alpha k + \beta$ the coefficients are $\alpha(m - i + 1) - 1$, independent of n (Table 3).

a_n	r_k	multiplier	staircase coefficients
$n!$	k	$n - m$	$m-1,\ m-2,\ \dots,\ 1,\ 0$
$(2n)!! = 2 \cdot 4 \cdots (2n)$	$2k$	$2(n-m)$	$2m-1,\ 2m-3,\ \dots,\ 3,\ 1$
$(2n-1)!! = 1 \cdot 3 \cdots (2n-1)$	$2k-1$	$2(n-m)-1$	$2m-1,\ 2m-3,\ \dots,\ 3,\ 1$
$1 \cdot 4 \cdot 7 \cdots (3n-2)$	$3k-2$	$3(n-m)-2$	$3m-1,\ 3m-4,\ \dots,\ 5,\ 2$
b^n	b	b	$-1,\ -1,\ \dots,\ -1$

Table 3. Proposition A.1 for some sequences. For powers b^n the growth factor does not increase, the coefficients are negative, and the identity reduces to the sum of a geometric series.

A.2. **A q-analogue.** An *inversion* of u is a pair of positions $i < j$ with $u_i > u_j$; let $\mathrm{inv}(u)$ be their number. With $[k]_q = 1 + q + \cdots + q^{k-1}$ (and $[0]_q = 0$) and $[k]_q! = [1]_q [2]_q \cdots [k]_q$, it is classical that $\sum_{u \in \mathfrak{S}_k} q^{\mathrm{inv}(u)} = [k]_q!$ [20, Cor. 1.3.13].

Proposition A.3 (cf. [20])**.** *For $0 \le m \le n-1$ let $e_i = \sum_{j=1}^{i-1} (m - j) = (i-1)m - \binom{i}{2}$. Then*

$$\sum_{u:\, w_m \le_{\mathrm{lex}} u} q^{\mathrm{inv}(u)} = [n-m]_q \Bigl[\sum_{i=1}^{m} q^{\,e_i + m - i + 1}\, [n-i]_q! + q^{\,e_{m+1}}\, [n-m-1]_q! \Bigr],$$

$$\sum_{u<_{\mathrm{lex}} w_m} q^{\mathrm{inv}(u)} = \sum_{i=1}^{m} q^{\,e_i}\,[m-i]_q\,[n-i]_q!,$$

and the two sums add up to $[n]_q!$. *For* $q=1$ *this is Proposition 3.3.*

Proof. Repeat the proof of Lemma 2.1, keeping track of inversions. Let $u \neq w_m$ first differ from w_m at position i. The first $i-1$ entries are $m, m-1, \ldots, m-i+2$; each has all smaller values to its right, so together they give e_i inversions. At position i the unused values are $\{1, \ldots, m-i+1\} \cup \{m+1, \ldots, n\}$, and choosing the r-th smallest ($r = 0, \ldots, n-i$) creates r inversions with later positions. The entry of w_m is the $(m-i)$-th smallest, so the smaller choices give $[m-i]_q$ and the larger ones $q^{m-i+1}[n-m]_q$. The remaining positions give $[n-i]_q!$. The permutations that agree with w_m in the first m positions give $q^{e_{m+1}}[n-m]_q! = q^{e_{m+1}}[n-m]_q[n-m-1]_q!$. □

So the multiplier $n-m$ becomes $[n-m]_q$: the $n-m$ larger choices at each step contribute consecutive numbers of inversions.

Appendix B. Computer verification and formal proofs

The files described here are embedded in this PDF as attachments (open the attachments panel of a PDF reader such as Adobe Acrobat to save them).

B.1. **What was checked, and how.** Table 4 lists every result with its computer check and, where there is one, its formal proof. The checks are in three files: `verify.py` (Python 3 with exact integer arithmetic, sympy for the q-analogue and mpmath for the Gamma function; its numbered blocks are cited as §1, §2, ...), `near_kurepa.c` and `near_kurepa.py`. The formal proofs are in the Lean 4 files `Basic.lean` and `Arithmetic.lean` (Lean and Mathlib v4.35.0). They contain no unproved steps (`sorry`), and `Axioms.lean` confirms that every theorem depends only on Lean's three standard axioms.

To avoid subtraction of natural numbers, the Lean files write $n = k+m+1$, so that the multiplier $n-m$ is $k+1$. For example, Proposition 3.1 reads:

```
def bracket (k m : ℕ) : ℕ := ∑ j ∈ range (m + 1), (k + j)!
def staircase (k m : ℕ) : ℕ := ∑ j ∈ range (m + 1), (j - 1) * (k + j)!

theorem staircase_identity (k m : ℕ) :
    (k + m + 1)! = (k + 1) * bracket k m + staircase k m
```

B.2. **Values of the shortfall.** The table lists $R_m(n) = n! - (n-m)\sum_{i=1}^{m+1}(n-i)!$. Its column $m=2$ is $(n-1)!$, its column $m=3$ is A052649 shifted by two ($R_3(n)$ is term $n-2$), and its diagonal $R_{n-1}(n)$ is term n of A065355. The columns $m \geq 4$ were not in the OEIS at the time of writing.

n	$m=1$	$m=2$	$m=3$	$m=4$	$m=5$	$m=6$
2	0					
3	0	2				
4	0	6	14			
5	0	24	54	86		
6	0	120	264	414	566	
7	0	720	1560	2424	3294	4166
8	0	5040	10800	16680	22584	28494
9	0	40320	85680	131760	177960	224184
10	0	362880	766080	1174320	1583280	1992360

Result	Computer check	Lean theorem (file)
Lemma 2.1	`verify.py` §2, via Prop. 3.3	not formalised
Prop. 3.1	§1: all $n < 40$, $0 \le m \le n-1$	`staircase_identity` (Basic)
Remark 3.2	§5: to 60 digits at real and complex points	`gamma_staircase` (Arithmetic)
Prop. 3.3	§2: rank for $n \le 9$; all of $\mathfrak{S}_n$ listed for $n \le 7$	not formalised
Prop. 4.1	§3: all $n \le 8$ and allowed (m, c)	identity only: `family_identity` (Basic)
Cor. 4.3	§3, case $m = n-1$	identity only: `left_factorial_endpoint` (Basic)
Equation (2)	used in §5c	`bracket_eq_factorial_mul_cofactor` (Basic)
Lemma 5.1	used in §5c	`ascFactorial_modEq` (Basic)
Prop. 5.2	§5c: primes $p < 30$, $n < 60$ (17700 cases)	`cofactor_modEq` (Basic), `cofactor_derangement` (Arithmetic)
Cor. 5.4	§5c: primes $p < 80$	`kellner`, `left_factorial_mod_prime`, `kurepa_iff` (Arithmetic)
Cor. 5.6	§5c: all $2 \le n < 60$	last claim only: `succ_dvd_cofactor` (Arithmetic)
Table 2	`near_kurepa.c`: odd $p < 2 \cdot 10^6$; `near_kurepa.py`: odd $p < 3 \cdot 10^4$ and $d_{11562} \bmod 11563$	not formalised
Prop. A.1	§5b: 400 random cases and Table 3	identity only: `mixed_radix_identity` (Basic)
Prop. A.3	§4: as polynomials for $n \le 7$; by listing permutations for $n \le 6$	not formalised

TABLE 4. Each result, its computer check, and its formal proof. "Identity only" means that the counting interpretation was checked by computer but not formalised.

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